

FORM TP 2006207



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MAY/JUNE 2006

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION
ENVIRONMENTAL SCIENCE

UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

PAPER 03/B

2 hours

11 MAY 2006 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **THREE** questions.
2. Answer **ALL** questions.
3. Write your answers in the answer booklet provided.
4. Graph paper is provided.
5. The use of a non-programmable calculator is allowed.

Answer ALL questions.

Table 1 presents population data for a Caribbean country for the period 1996 to 2000. At the beginning of 1996, the population size of the country was 2,500,000. The data in Table 1 represents changes in the population at the end of each year. Use the data in Table 1 to answer Questions 1 and 2.

TABLE 1: POPULATION DATA FOR A CARIBBEAN COUNTRY, 1996 - 2000

Population parameters	Year				
	1996	1997	1998	1999	2000
Number of births	60,200	59,720	59,220	59,160	58,640
Number of deaths	10,000	8,500	7,000	5,900	4,500
Number immigrated	300	400	300	200	500
Number emigrated	500	620	500	400	700

1. (a) Calculate, showing each step,
 - (i) the change in the size of the population at the end of each year [5 marks]
 - (ii) the population size at the end of each year [5 marks]
 - (iii) the average natural rate of increase of the population [7 marks]
 - (iv) the birth rate and the death rate of the population at the start of the period [4 marks]
 - (v) the birth rate and the death rate of the population at the end of the period. [4 marks]
 - (b) Predict the size of the population in 2050. [5 marks]
- Total 30 marks**
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2. (a) Use a bar chart to represent the population of the country over the five-year period. [12 marks]
 - (b) Suggest reasons for the rate of growth of the population during the period. [8 marks]
 - (c) Discuss an appropriate programme that could be implemented to control the growth rate of the population. [10 marks]

Total 30 marks

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3. Outline the MOST appropriate sampling technique for EACH of the following:

- (a) To investigate the transition from one community to another [8 marks]
- (b) To estimate population density and percentage cover [10 marks]
- (c) To estimate population size of mobile (moving) organisms [7 marks]
- (d) To investigate stationary organisms [5 marks]

Total 30 marks

END OF TEST